

Usability of Biometrics Authentication in ATMs

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1 Abstract

This research explores the transition from traditional PIN-based authentication to biometric methods in Automated Teller Machines (ATMs), driven by increasing security concerns. The study evaluates user perspectives on the usability and security of biometric technologies, focusing on fingerprint and facial recognition. Our survey was conducted to gauge their familiarity, experiences, security concerns, and preferences regarding these technologies. The findings reveal widespread familiarity with biometrics and a general ease of use, with a notable preference for fingerprint scanning over facial recognition. Despite this, substantial privacy and security concerns remain, particularly regarding data breaches and the storage of biometric data. Nonetheless, a majority of participants showed a willingness to adopt biometric authentication. These results indicate that while biometric technologies offer significant potential for improving ATM security, addressing user concerns about data protection is critical for broader acceptance. The study underscores the necessity for the financial industry to enhance security protocols and ensure greater transparency to foster user trust in biometric systems.

2 Problem and Background

In response to the increasing security risks associated with PIN-based authentication methods, the banking industry is witnessing a notable shift towards adopting biometric technology for user verification at ATMs (Automated Teller Machines). This transition is motivated by the vulnerabilities inherent to PINs, such as forgetfulness and potential leakage.

Furthermore, strict regulatory mandates, such as the United States' Know Your Customer (KYC) [5] regulation determine the necessity for a thorough understanding of clients and the risks involved in the banking industry.

Amid a landscape where technology consistently reshapes notions of convenience and safety, the banking industry stands at the vanguard of this transformative process. Biometric ATMs, representing an innovative breakthrough, are poised to revolutionize the landscape of financial transactions.

A Biometric ATM constitutes a modern iteration of an automated teller machine, utilizing biometric authentication methods for user verification based on physical or behavioral characteristics. This approach ensures a secure, personalized, and efficient banking experience. Unlike conventional ATMs that require a card and PIN for user verification, biometric ATMs utilize biometric technologies such as fingerprint recognition, facial recognition, iris scanning, and voice recognition to verify a customer's identity, which facilitates access to many ATM services [2], including cash withdrawals, deposits, balance inquiries, fund transfers, bill payments, and more advanced financial services like loan applications and account management.

Biometric technology enhances user convenience by eliminating the need to remember PINs. Moreover, it fortifies security measures as it is harder for malicious actors to replicate fingerprints or counterfeit a victim's face to circumvent the technology. Although biometric technology presents a promising solution for the challenges experienced by banks, it is imperative to address the pivotal security concerns expressed by users to expedite the widespread embrace of this transformative technology.

The primary objective of this user study is to gain insights into users' perceptions regarding the Usability and Security implications of integrating Biometric into bank ATMs (Automated Teller Machines). We intend to assess users' comfort level using biometric technology as an authentication method at ATMs. The study will focus on understanding users' famil-

ilarity with biometric technologies in their daily lives, their overall experiences with these technologies, and encountered issues. Additionally, we aim to gather information about any privacy or security concerns associated with Biometric technology that users may harbor.

To better understand user attitudes toward biometric security methods, we are conducting a study to identify potential obstacles to adopting biometric technology in ATMs in the coming years. Additionally, as part of this study, we explore why users prefer fingerprint scanning over face recognition to gain insights into user perspectives on specific biometric technologies. In summary, this study aims to analyze user perceptions of the usability and security of biometric technologies, particularly in the context of ATMs.

Our user study is driven by three core research questions:

RQ1: What is the users' perception of the Usability & Security of using Biometric in ATMs?

RQ2: What could be potential barriers to adopting Biometric in ATMs?

RQ3: Do users prefer fingerprint scanning over face recognition, and if so, why?

3 Related Work

The global adoption of biometric ATMs is becoming increasingly popular. In some countries, it is introduced as an additional layer of security to enhance overall security. In others, it is the sole authentication, replacing PIN and card-based authentication in ATMs. Biometric usage makes the authentication process at ATMs more convenient.

For instance, Seven Bank in Japan allows customers to access ATM services such as cash deposits and withdrawals using facial recognition technology at around 20,000 ATMs in Seven-Eleven stores. Approximately 13,000, or half, of all Seven Bank ATMs already have face biometric technology installed. [4] This implementation aims to improve security and make banking services more user-friendly and accessible.

South Africa's First National Bank has begun fingerprint biometric identification and verification [6] at its branches, thereby promoting Biometric usage in the banking sector. Caxia Bank in Spain deployed ATMs with facial recognition technology amid Covid-19 with the goal to offer better user experience and greater security [3] during transactions.

In this area, several case studies have been conducted to explore the use of biometric-based fingerprint verification systems for ATM machines [8]. One study implemented this system to enhance security and provide convenience to customers by eliminating the need for cards and PINs. Another

study reviewed the benefits of fingerprint-based biometric systems in ATM banking [1], focusing on increased client confidence and improved security. Additionally, a research framework was developed to study the adoption of biometric ATM authentication in Kenyan financial institutions [7], identifying performance expectancy, effort expectancy, social influence, and user privacy as critical factors influencing its acceptance and use. These research studies compelled us to investigate whether users would prefer fingerprint scanning over facial recognition or other biometric technologies.

One additional pertinent case study we encountered relates to our research focus on biometrics in banking security [9]. The study delves into the challenges, opportunities, and positive attributes associated with biometrics, examining the integration of biometric solutions in banking institutions through an exploration of technological, managerial, social, and ethical considerations. This evaluation identifies key success factors that could facilitate the deployment of biometric authentication systems in banking establishments, particularly within a prominent New Zealand-based bank.

4 Methodology

In this section, we describe how we designed our study, including the types of questions we asked participants, our recruitment strategy, data collection methods, and the method used for analyzing the data.

4.1 Surveys

The recruitment strategy involved conducting a survey to gain insights into the users' perspectives on biometric concerning usability and security, familiarity, potential barriers they could face in adopting biometric in ATMs, the type of biometric authentication they would prefer, and lastly their willingness to adopt the biometric over traditional ATM PIN-based system.

The survey's focus is to understand the users' point of view on using biometric in ATMs over traditional ATM cards and PIN-based systems as well as the usability, security, and privacy concerns of biometric systems. The survey was conducted on the LimeSurvey application and our recruitment strategy was to reach out to as many participants as possible by personally sending out the survey link through.

The survey is divided into four sections:

1) Familiarity with biometric technology usage: to begin with the survey, it is essential to assess the fundamental understanding participants possess regarding biometric technology. This includes determining whether they have previously engaged with any forms of biometric systems, such as facial recognition or fingerprint scanning.

2) Experience with biometric authentication: The experience participants must have/had and the ease or difficulty of using biometric technology.

3) Security and privacy concerns associated with biometric technology: The kind of potential issues related to security and privacy users expect while dealing with biometric authentication methods such as fingerprint scanning, face recognition, etc.

4) User perceptions and preferences: this section deals with questions regarding participants' preferences between ATM PIN-based systems and biometric authentication for future use, along with the rationale behind their choices.

To get better insights into the possibilities of future improvement of biometric authentication, a comprehensive survey was conducted. The survey consisted of 16 questions. The questions were a mix of yes/no questions, multiple-choice with comments, and Likert-scale type questions to assess the degree of satisfaction/dissatisfaction, and agreement/disagreement. Participants were given 10-12 minutes to fill out the survey anonymously. In total, 54 participants responded, but not all surveys were fully completed. Due to this, 45 survey results were included in the analysis.

4.2 Analysis

Upon data collection, quantitative questions were analyzed using the LimeSurvey statistics tool, while qualitative questions were examined with QCMap software.

a) Quantitative questions: the LimeSurvey tool automatically generated the statistics from the survey data. Additionally, the tool offers numerous options to further develop the survey results into various charts or graphs of your choice.

b) Qualitative questions: there were in total 5 qualitative questions. The responses were analyzed with the help of QCMap software.

Let us examine the steps involved in conducting qualitative analysis using QCMap.

1) The responses have to be stored in a document file.

2) After defining the research question, we have to set certain rules before beginning with the coding and the analysis part.

3) Choose a content analytical technique. According to our survey data, the preferred category would be inductive category formation. It is a bottom-up approach. In this approach, we get the narrative from the raw data and the codes are created with no preconceived notion.

4) To continue with inductive category formation, certain sets of rules are to be set such as the Content Analytical Units section consisting of coding units, context units, and recording

units (number of documents). The selection criterion and abstraction level have to be defined as well.

5) After setting the rules, the coding can be done. In the coding, the raw data has to be coded in the categories by selecting the text. The tool will list the categories in the category system. Must ensure that the categories defined fit the research question. After finalizing the coding part, we can move on to the analysis. A frequency analysis of the category occurrences is shown. The user can view and download the results.

To understand the users' perception of the biometric authentication used at ATMs, conducting a survey proves to be an effective method. It allows for the formulation of a well-structured set of questions directed at individuals for their responses. Additionally, surveys are straightforward to administer and facilitate the derivation of clear results.

5 Results and Discussion

5.1 Question Group 1: Familiarity with Biometric Technology Usage

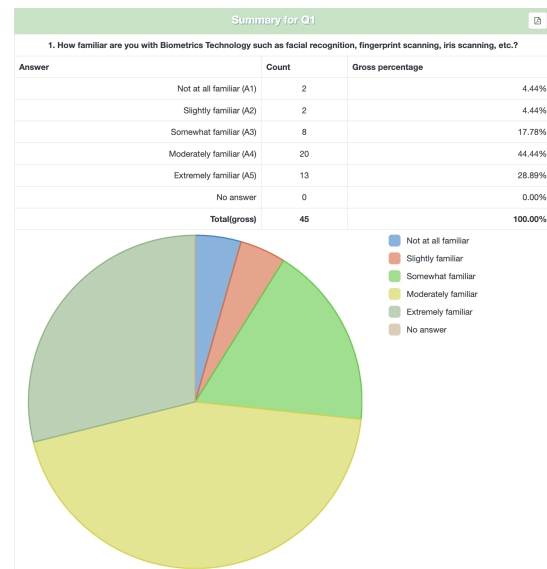


Figure 1: Level of familiarity with Biometric

Figure 1 shows that only 2 out of 45 participants (4.44%) were not at all familiar with biometric, while 95.56% of people have used biometric in their daily lives. This emphasizes that most of the people are familiar with the emerging biometric technology.

Figure 2 illustrates around 91.11% of the participants have used fingerprint scanning and only 8.89% of the participants have used facial recognition to unlock their mobile phones or any applications.

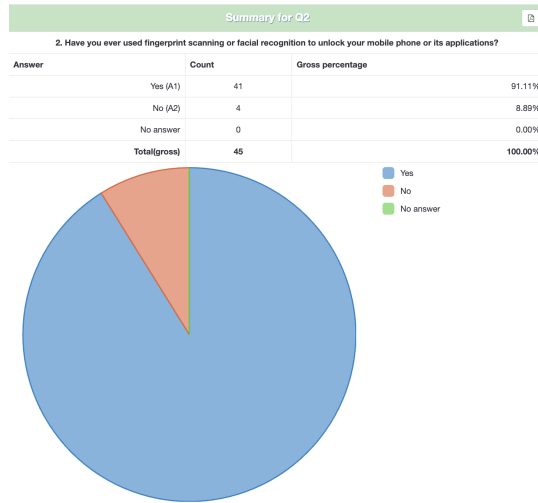


Figure 2: Biometric usage to unlock mobiles or applications

Figure 3 indicates that fingerprint scanning is the most commonly used biometric among the survey participants, followed by facial recognition, then voice recognition, and lastly iris scanning. The results of this question raised questions about why participants favored fingerprint scanning over facial recognition. Therefore, we used QCAMap to conduct further analysis.

Figure 4 demonstrates that 51.11% of the participants use ATMs on a monthly basis. This shows that ATMs are still frequently used. Implementing biometric authentication in ATMs will benefit over 50% of bank users, and the survey results from this study could also contribute to this.

Summarizing the results related to the 'Familiarity with Biometric Technology Usage' question group of our survey, the survey questions from this group aimed to determine the extent to which the participants were familiar with biometric technologies such as facial recognition, fingerprint scanning, voice recognition and iris scanning. The results indicate that most participants had a strong familiarity with biometric technologies. Only a small fraction of the respondents expressed a lack of familiarity. This high level of familiarity and awareness may be due to the increasing integration of biometric technologies in consumer devices, such as smartphones and laptops. The widespread familiarity suggests that people are ready to embrace biometric authentication in a broader sense, which could boost the adoption of biometric ATMs. For banks and other financial organizations wishing to introduce or increase biometric authentication solutions, this knowledge could be crucial.



Figure 3: Type of biometric used

5.2 Question Group 2: Experience with Biometric Authentication

Figures 5 and 6 shows that all the participants found using biometric ATMs very easy. Not a single participant had any difficulty using biometric ATMs. Although there may be some bias in the results, we can confidently say that the majority of bank users will find biometric authentication easy and convenient to use in ATMs.

Figure 7 shows that out of 45 participants, approximately 34 individuals encountered problems with biometrics failing to recognize their face or fingerprint. This suggests that there is a significant risk of false negatives in biometric identification, which should be taken into account when considering implementing biometrics for identification purposes at ATMs.

Figures 8 and 9 indicates that fingerprint scanning is the most favored biometric authentication method. We also thoroughly examined the qualitative data to understand the reasons behind this preference as part of our user study. The findings have been included here 5.5.

Summarizing the question group 'Experience with Biometric Authentication', as part of this group we wanted to explore the practical application of biometric technology so we asked the participants whether they have used fingerprint scanning or facial recognition to unlock their mobile phones or its applications. The responses provided a clear picture of how experienced people were with biometric authentication



Figure 4: Frequency of ATM usage

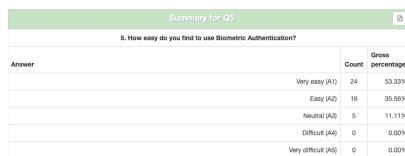


Figure 5: Ease of use of biometrics

in their day-to-day lives. These findings reflect that the majority of the participants have substantial exposure to biometric technologies. This also suggests that most individuals are not just aware but are comfortable with biometric authentication, reflecting that similar technologies can be adopted in other aspects, like banking. The high rate of usage also implies that people are comfortable with the operational and security aspects of biometrics, which might reduce resistance to the adoption of biometric in more vital cases like ATMs. Understanding this aspect could be crucial for banks when considering the implementation of biometric authentication systems in their services, as it helps them realize the receptiveness to new biometric security measures by consumers.

5.3 Question Group 3: Security and Privacy Concerns associated with Biometric Technology

While assessing the willingness to use biometric authentication at ATMs as shown in Figure 10 and Figure 11, only 35.56% of participants expressed readiness to use their fingerprint or facial recognition for withdrawals, while 64.44% were opposed.

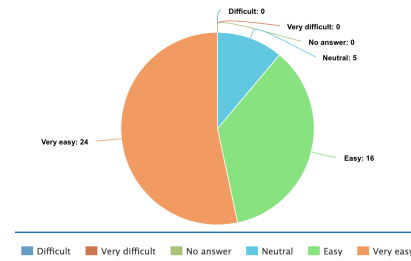


Figure 6: Ease of use of biometrics



Figure 7: False negatives using biometric for identification

Evaluating the safety of banks holding biometric information as portrayed in Figure 12 and Figure 13, about 51% of respondents felt safe to varying degrees, showing some trust in banks' handling of biometric data. Conversely, nearly half expressed concerns, with 31.11% feeling not safe and 17.78% somewhat unsafe, due to fears of data breaches and the potential for biometric exploitation.

When asked about concerns on potential data breaches, a significant majority expressed worry as shown in Figure 14: 60% are extremely concerned, 20% are moderately concerned, and 11.11% are somewhat concerned about data breaches, totaling over 91% indicating varying levels of concern. Only a small fraction, 8.89%, reported being slightly concerned, and notably, none of the respondents felt completely unconcerned.

In the survey question about the safety of biometric authentication compared to PIN-based systems, opinions were divided among the participants as depicted in Figure 15: 11.11% strongly agree and 20% agree that biometric are safer, totaling 31.11% in favor. However, an equal percentage (20%) disagree, and 17.78% strongly disagree, indicating skepticism

Summary for Q7			
7. Which method would you prefer for Biometric Authentication?			
Answer	Count	Gross percentage	
Fingerprint Scanning (A1)	27	60.00%	
Face Recognition (A2)	18	40.00%	

Figure 8: Most preferred type of biometrics

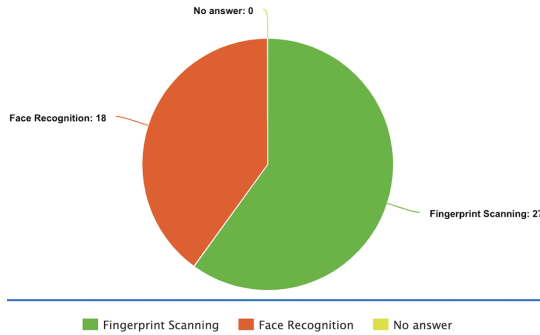


Figure 9: Most preferred type of biometrics

towards biometric security.

Summarizing the question group 'Security and Privacy Concerns associated with Biometric Technology', this question group highlights a paradox in user perception towards biometric technologies: while familiarity and personal use are high, significant concerns about security and privacy persist, especially regarding the use of biometric in sensitive areas like ATM transactions. Over 91% of users are worried about data breaches, and a majority remain skeptical about the comparative security of biometric versus traditional methods. This underscores the need for financial institutions to bolster security measures, enhance transparency, and educate consumers about biometric systems to bridge the gap between comfort with personal devices and trust in biometric security for financial applications.

5.4 Question Group 4: User perceptions and preferences

Summary for Q8			
8. Would you be willing to use your fingerprint/face to withdraw cash at an ATM?			
Answer	Count	Gross percentage	
Yes (A1)	16	35.56%	
No (A2)	29	64.44%	

Figure 10: Willingness to use biometric authentication at ATMs

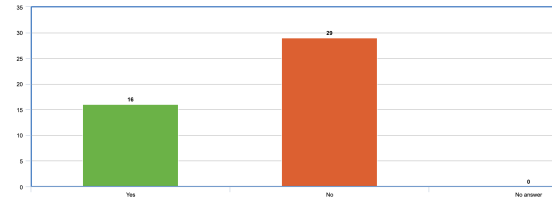


Figure 11: Willingness to use biometric authentication at ATMs

Summary for Q9			
9. How safe would you feel if your bank had your Biometric information?			
Answer	Count	Gross percentage	
Very safe (A1)	3	6.67%	
Somewhat safe (A2)	6	13.33%	
Safe (A3)	14	31.11%	
Somewhat unsafe (A4)	8	17.78%	
Not safe (A5)	14	31.11%	

Figure 12: Safety of banks holding biometric information

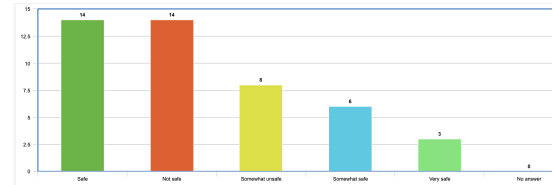


Figure 13: Safety of banks holding biometric information

Summary for Q10			
10. How concerned are you about a potential data breach? (Data Breach: A data breach happens when personal information is accessed or disclosed without authorization, or is lost.)			
Answer	Count	Gross percentage	
Extremely concerned (A1)	27	60.00%	
Moderately concerned (A2)	9	20.00%	
Somewhat concerned (A3)	6	13.33%	
Slightly concerned (A4)	4	8.89%	
Not at all concerned (A5)	0	0.00%	
No answer	0	0.00%	
Total (n)	46	100.00%	

Figure 14: Concerns on potential data breaches

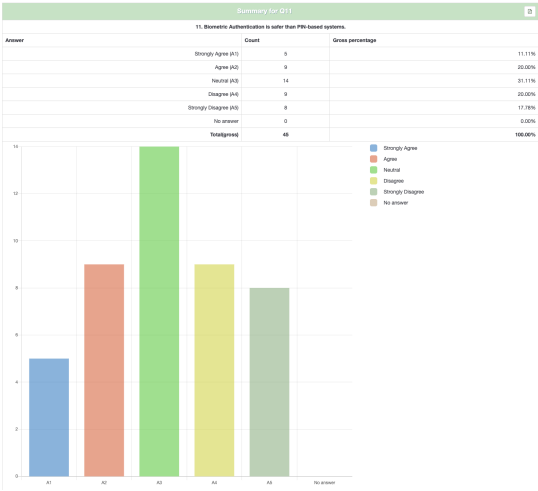


Figure 15: Safety of biometric authentication compared to PIN-based systems

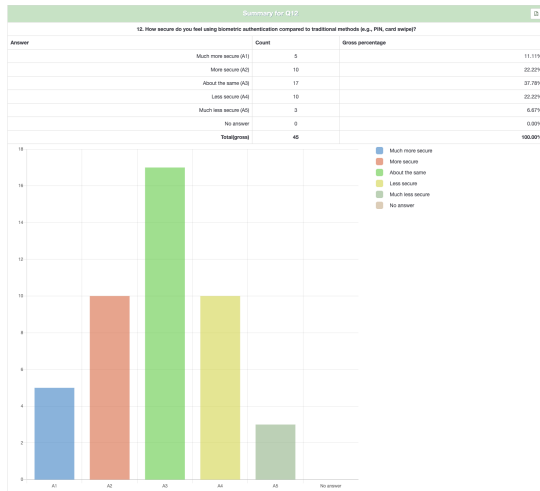


Figure 16: User perceptions on security of biometric compared to traditional methods

Assessing how secure participants feel using biometric authentication compared to traditional methods like PINs and card swipes, responses were varied as depicted in Figure 16: 11.11% feel it is much more secure, and 22.22% feel it is more secure. However, an equal percentage (22.22%) feel it is less secure, and 6.67% believe it is much less secure. The largest group, at 37.78%, feels biometric authentication offers about the same level of security as traditional methods.

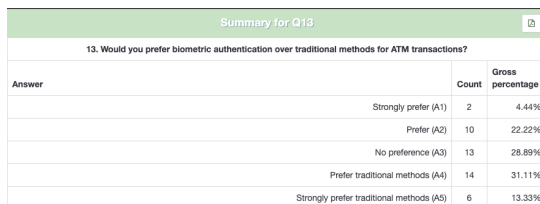


Figure 17: Users' preference for biometric authentication over traditional methods

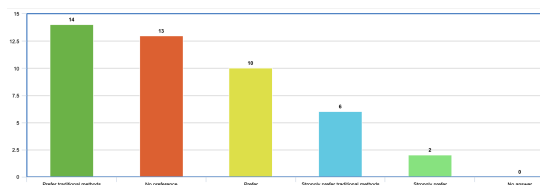


Figure 18: Users' preference for biometric authentication over traditional methods

While assessing preference for biometric authentication over traditional methods for ATM transactions, responses varied as illustrated in Figure 17 and Figure 18: 26.66% of participants prefer or strongly prefer biometric, while a higher

percentage, 44.44%, prefer or strongly prefer traditional methods such as PINs and card swipes. The largest single group, making up 28.89%, expressed no preference between the two methods.

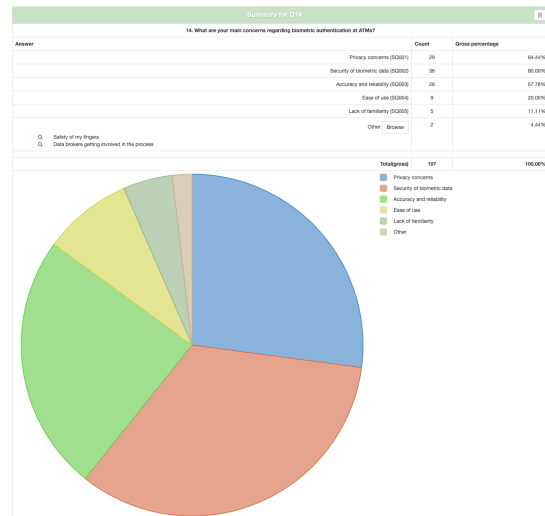


Figure 19: Main concerns regarding biometric authentication at ATMs

Querying main concerns regarding biometric authentication at ATMs as shown in Figure 19, security of biometric data emerged as the most significant worry with 80.00% of participants highlighting it. Privacy concerns were also substantial, noted by 64.44% of respondents. Accuracy and reliability were mentioned by 57.78%, while ease of use and lack of familiarity with biometric systems were less pronounced concerns, cited by 20.00% and 11.11% respectively. A small fraction (4.44%) indicated other concerns.

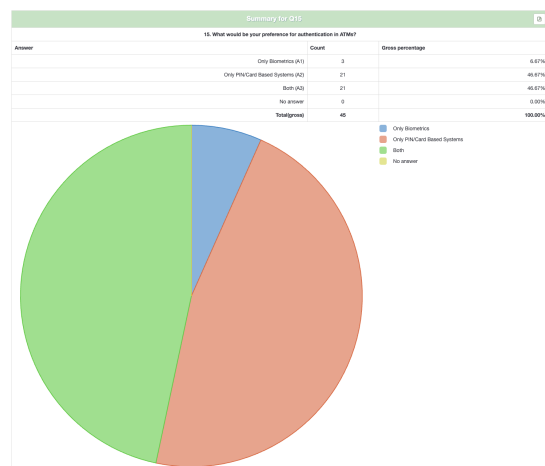


Figure 20: Preferred authentication methods at ATMs

In response to the survey question about preferred authentication methods at ATMs as depicted in Figure 20, respondents displayed a split preference for either traditional PIN/card-based systems or a combination of biometric with traditional methods, each garnering 46.67% of the total responses. Only a small minority, 6.67%, favored using only biometric for authentication.

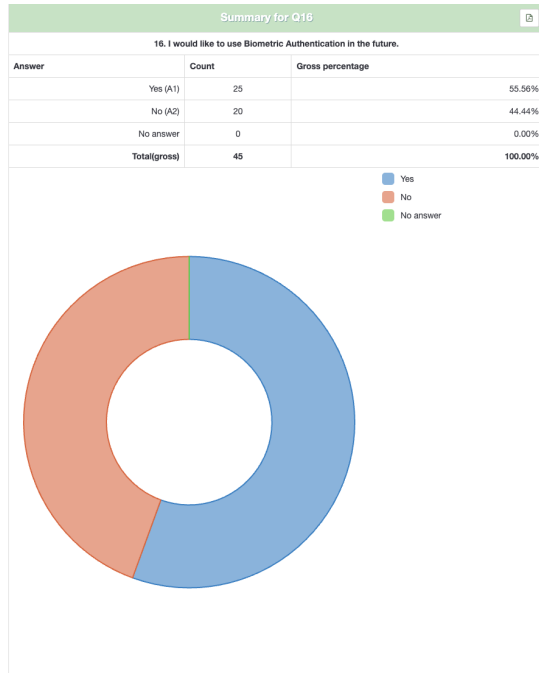


Figure 21: Security and privacy concerns with biometric

In response to the survey question about willingness to use biometric authentication in the future as illustrated in Figure 21, a slight majority of participants, 55.56%, expressed interest in adopting biometric methods. However, a significant minority, 44.44%, are still hesitant, indicating no interest in using biometric authentication going forward.

Summarizing the results from the question group 'User perceptions and preferences', indicate a complex landscape of user perception towards biometric authentication at ATMs. While there is an openness to integrating biometric features, a significant portion of users prefer the security and familiarity of traditional PIN and card methods or a combination of both. Security and privacy concerns dominate, underlining the need for robust protective measures and transparency in handling biometric data. Moving forward, the financial and technology sectors must address these apprehensions and demonstrate the reliability and benefits of biometric to encourage wider acceptance and confidence among users.

5.5 Qualitative Results

A total of five qualitative questions were included in the survey.

Research Question 1: How easy do you find it to use Biometric Authentication?

After deducing the survey answers, it is divided into three main categories:

Category - 1: Easy to use and convenient

Response: No need to remember the password, Fairly easy for users.

Category - 2: No proper feedback

Response: do feel the scanning part is trickery

Category - 3: Biased with password usage

Response: I'm happy just using a password instead

Research Question 2: Which method would you prefer for Biometric Authentication? After deducing the survey answers, it is divided into seven categories:

Category-1: Direct contact in biometric

Response: Feels more safe and reliable as it requires direct contact/touch

Category-2: False negatives in biometric

Response: For fingerprints you have to make sure you have dry hands and the scanner is clean enough to pick up your prints

Category-3: Issue with face data storage

Response: I think I would not want my face information to be stored anywhere so I would opt for fingerprint

Category-4: Fingerprint is more reliable

Response: It's not dependent on lightning or attire.

Category-5: Seasonal issues with fingerprint

Response: Fingerprint sometimes cause trouble in winter.

Category-6: Face recognition has lesser false negatives

Response: Face recognition is more accurate.

Category-7: Lighting issues with face recognition

Response: At night in direct sunlight or when e.g. my face is obstructed by a face mask face recognition does not work as well.

Research Question 3: Would you be willing to use your fingerprint-face to withdraw cash at an ATM?

After deducing the survey answers, it is divided into seven categories:

Category-1: Biometric data theft at ATMs

Response: It's a publicly accessed machine to which any

tampering is relatively easier and because of that I won't feel comfortable.

Category-2: Expects additional layer of security

Response: I do feel that standalone biometric is not sophisticated enough to prevent the breach

Category-3: Danger to life (biometric data theft)

Response: I don't want my fingers to be cut

Category-4: Concern with biometric data storage at bank

Response: I don't feel secure knowing that my biometric information is with bank

Category-5: Low severity issues with PINs

Response: My card, phone app and one time code requests work fine and flawlessly to use a ATM

Category-6: Biometric is safer than PINs

Response: Yes since it seems safer than just a 4 digit pin.

Category-7: Facial recognition over other biometric

Response: Only if it was facial recognition

Research Question 4: How safe would you feel if your bank had your Biometric Information?

After deducing the survey answers, it is divided into four categories:

Category-1: Biometric data accesible in multiple banks

Response: Biometric data accesible in multiple banks; Having multiple bank accounts, I wouldn't like all the entities to have my biometric data.

Category-2: Biometric data leak

Response: If it gets leaked, that would be really scary

Category-3: Trust bank with biometric data

Response: I would trust no company with my biometric data.

Category-4: Banks must assure transparency and total security

Response: Proper Documentation should be provided regarding the Data Protection Laws.

Research Question 5: Would you prefer biometric authentication over traditional methods for ATM transactions?

After deducing the survey answers, it is divided into four categories:

Category-1: Permanence of biometric data

Response: I want my fingers to stay intact

Category-2: Biometric as additional security

Response: Increased safety and reliability as well as retaining biometric as fallback identification

Category-3: PINs are less safe

Response: I like both but PINs seem less safe to me

Category-4: Okay with using PINs

Response: Even if there's a chance for brute-forcing PIN, I still prefer using a method like that for secure transactions

Category-5: Biometric is convenient

Response: Don't have to remember passwords or worry about card loss

6 Conclusion

In conclusion, our research revealed that, contrary to our initial assumptions, more than half of the people we surveyed were already familiar with biometric technology in their daily lives, whether through their work or mobile phones, and found biometric usage to be more convenient. Another interesting finding was that users were more inclined to use fingerprint recognition compared to facial recognition.

However, a significant number of participants expressed that biometric technology was not their preferred banking technology. They cited privacy and security concerns, including worries about data breaches, as the main reasons for their reluctance. Almost half of the participants expressed concerns about the storage of their biometric data at banks for authentication purposes. These outcomes were expected, considering the potential for personal data to be compromised, as fingerprint and facial data are more sensitive than randomly generated PINs. Participants raised concerns about biometric data storage security, and privacy, as well as the accuracy and reliability of the technology, particularly with false positives and false negatives.

In summary, customers have more to lose when using biometric technology, while banks primarily benefit from the ease of authentication. Based on the data we collected, although more than half of the participants are willing to use biometric as a method of authentication in ATMs, they still have security concerns. Therefore, we strongly recommend that the financial sector focus on enhancing the security of banking information rather than prioritizing ease of use.

7 Appendix

Note: We have made use of Generative AI from Grammarly to improve our writing, that is to make it more sound more academic rather than layman and our grammatical mistakes. We have also used ChatGPT to summarize few points.

8 Survey Questions

1. How familiar are you with Biometrics Technology such as facial recognition, fingerprint scanning, iris scanning, etc.?
2. Have you ever used fingerprint scanning or facial recognition to unlock your mobile phone or its applications?
3. Which type of Biometric Authentication have you used?
4. How often do you use ATMs?

Experience with Biometric Authentication

5. How easy do you find to use Biometric Authentication?
6. While using Biometric Authentication, did you encounter a situation where the system did not recognize your face or fingerprint?
7. Which method would you prefer for Biometric Authentication?

Security and Privacy Concerns Associated with Biometrics Technology

8. Would you be willing to use your fingerprint/face to withdraw cash at an ATM?
9. How safe would you feel if your bank had your Biometric Information?
10. How concerned are you about a potential data breach? (Data Breach: A data breach happens when personal information is accessed or disclosed without authorization, or is lost.)
11. Biometric Authentication is safer than PIN-based systems.

User Perceptions and Preferences

12. How secure do you feel using biometric authentication compared to traditional methods (e.g., PIN, card swipe)?

13. Would you prefer biometric authentication over traditional methods for ATM transactions?

14. What are your main concerns regarding biometric authentication at ATMs?

15. What would be your preference for authentication in ATMs?

16. I would like to use Biometric Authentication in the future.

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